

From Pairs to Portfolios: A Guide to Cointegration and Statistical Arbitrage

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1 Introduction

In the current landscape of quantitative finance, the distinction between correlation and cointegration is frequently misunderstood, leading to fragile trading strategies that fail when markets turn volatile. While many aspiring quants look for assets that simply "move together," this superficial relationship often breaks down exactly when reliability is needed most.

This article aims to bridge the gap between econometric theory and practical algorithmic trading. We will move beyond the basic technical analysis often found in retail trading blogs to offer a rigorous, academic approach to market neutrality. Our journey will take us from the theoretical foundations of simple pairs trading to the complex implementation of basket strategies using Vector Error Correction Models (VECMs), culminating in a practical application on the WorldQuant Brain platform.

1.1 The Drunk and the Dog: An Intuitive Framework

Before diving into the algebra of stationarity, it is crucial to dispel the myth that assets moving together are necessarily linked. To visualise the difference between a spurious correlation and a true economic relationship, we can utilise the classic "Drunk and the Dog" analogy.

Imagine a drunk man walking out of a bar. He stumbles randomly down the street. Now, imagine a stray dog wandering down the same street. Occasionally, they might both turn left or right at the same time. If you plotted their paths, you might calculate a high correlation coefficient—they are moving in the same direction by chance. However, this relationship is useless for prediction; if the drunk turns left, there is no guarantee the dog will follow. This represents correlation.

Now, imagine the drunk has his dog on a leash. The drunk walks in a random, stochastic path, and the dog wanders excitedly in various directions to sniff trees or fire hydrants. The distance between them expands and contracts, but they can never drift entirely apart because the leash—the fundamental economic tether—keeps them bound together.

In financial terms, the drunk and the dog are cointegrated. While their individual price paths (random walks) are unpredictable, the distance between them (the spread) is not. If the dog wanders too far away (the spread widens), the leash's tension eventually pulls him back toward the drunk (mean reversion).

This concept forms the bedrock of Statistical Arbitrage. We are not betting on where the drunk is going; we are betting that, eventually, the leash will pull the dog back.

2 Theoretical Foundation: Engle-Granger Method

With the intuition of the "Drunk and the Dog" established, we must translate this analogy into a rigorous statistical framework. The "leash" that binds our two assets together is mathematically defined by the concept of stationarity.

To identify this relationship, we turn to the Engle-Granger two-step method. While simple compared to modern multivariate methods, it remains the standard entry point for understanding the mechanics of pairs trading.

2.1 Regression

The first step is to identify the linear relationship between the two asset price series, P_t^A and P_t^B (representing the drunk and the dog). We regress one asset against the other to isolate the residuals [[1]].

We start with the linear model:

$$P_t^A = \alpha + \beta P_t^B + \epsilon_t$$

Here, β represents our Hedge Ratio—the amount of asset B required to hedge a unit of asset A. To derive the optimal β that minimises the variance of the residuals (the spread), we employ Ordinary Least Squares (OLS). Mathematically, we aim to find the estimator $\hat{\beta}$ that minimises the Sum of Squared Residuals (SSR):

$$SSR = \sum_{t=1}^T \left(P_t^A - \hat{\alpha} - \hat{\beta} P_t^B \right)^2$$

By taking the partial derivative with respect to β and setting it to zero, we arrive at the closed-form solution for the hedge ratio:

$$\hat{\beta} = \frac{\sum_{t=1}^T (x_t - \bar{x})(y_t - \bar{y})}{\sum_{t=1}^T (x_t - \bar{x})^2} = \frac{\text{Cov}(P_t^B, P_t^A)}{\text{Var}(P_t^B)}$$

Once we have derived the optimal hedge ratio ($\hat{\beta}$), calculating the intercept is straightforward. It is the difference between the mean of the dependent variable and the mean of the independent variable weighted by our beta:

$$\hat{\alpha} = \overline{P^A} - \hat{\beta} \overline{P^B}$$

Where: $\overline{P^A}$ is the sample mean of the prices of Asset A. $\overline{P^B}$ is the sample mean of the prices of Asset B.

This calculation ensures that the resulting spread (ϵ_t) is as "tight" as possible. Rearranging the original equation gives us the tradable portfolio:

$$\epsilon_t = P_t^A - \hat{\beta} P_t^B - \hat{\alpha}$$

2.2 Testing for Stationary

Generating a spread is easy; proving it is mean-reverting is difficult. For a pairs trade to be viable, the residual series ϵ_t must be stationary, meaning the spread fluctuates around a constant mean rather than drifting indefinitely.

To confirm this, we apply the Augmented Dickey-Fuller (ADF) test. While many traders simply look for a low p-value, understanding the underlying mechanism is crucial for avoiding false positives.

The ADF test evaluates whether the spread follows a "Unit Root" process (a random walk). This means:

1. If residuals follow the unit root process, it is not stationary, and shocks have permanent effects.
2. If residuals don't follow a unit root process, they are stationary.

Hence, if stationary, the residuals have constant mean and variance, confirming that cointegration exists between the assets. We estimate the following regression on the residuals ϵ_t :

$$\Delta\epsilon_t = \alpha + \gamma\epsilon_{t-1} + \sum_{i=1}^p \delta_i \Delta\epsilon_{t-i} + u_t$$

There are two critical components here:

1. The Lagged Level ($\gamma\epsilon_{t-1}$): This is the heart of the test. We test the Null Hypothesis $H_0 : \gamma = 0$.
 - If $\gamma = 0$, the current change $\Delta\epsilon_t$ does not depend on the previous level. Shocks to the spread are permanent (the dog runs away).
 - If $\gamma < 0$, the spread corrects itself. The farther ϵ_{t-1} drifts from zero, the stronger the pull of $\Delta\epsilon_t$ in the opposite direction (the leash pulls back)
2. The Augmentation Terms ($\sum \delta_i \Delta\epsilon_{t-i}$): These lagged difference terms are added to account for serial correlation in the error term, ensuring the validity of the test statistics.

Crucially, under the null hypothesis of a unit root, the t-statistic for γ ($\hat{t}_\gamma$) does not follow the standard Student's t-distribution. Because the regressor ϵ_{t-1} is itself non-stationary under the null, the limiting distribution of the test statistic is the non-standard Dickey-Fuller distribution, which is skewed to the left. Consequently, standard critical values (e.g., -1.96 for 5% significance) are invalid and would lead to frequent false rejections (Type I errors). Instead, we must compare our calculated t-statistic against the specific critical values simulated and tabulated by MacKinnon (1996)¹. If the t-statistic is more negative than these critical values (typically around -3.43 for a 1% significance level), we reject the null hypothesis, mathematically confirming that the relationship is not spurious and a true arbitrage opportunity exists [[3]].

2.3 Application of Engle-Granger Using Python

Having established the theoretical framework of the Engle-Granger two-step method, we now turn to a practical implementation using Python. The following script utilises the `yfinance` library for

data acquisition and statsmodels to perform the cointegration tests. We specifically employ the `coint` function, which automates the regression and subsequent stationarity test on the residuals. This allows us to empirically validate the 'drunk and dog' hypothesis between our chosen pair, Goldman Sachs (GS) and Morgan Stanley (MS), to confirm if a statistically significant tether exists [[4]].

```

1 # Necessary libraries
2 import yfinance as yf
3 import pandas as pd
4 import numpy as np
5 import statsmodels.api as sm
6 from statsmodels.tsa.stattools import coint
7
8
9 # Helper function to download and clean data
10 def get_prices(tickers, start_date='2021-01-01', end_date='2025-12-01'):
11     data = yf.download(tickers, start=start_date, end=end_date)['Close']
12     data = data.dropna() # Remove missing values
13     return data
14
15 # =====
16 Engle-Granger Test (2 Stocks)
17 # =====
18
19 # Let's test two Banks: Goldman Sachs (GS) and Morgan Stanley (MS)
20 pair_tickers = ['GS', 'MS']
21 df_pair = get_prices(pair_tickers)
22
23 # The 'coint' function runs the Engle-Granger two-step process
24 # It returns the t-statistic, p-value, and critical values
25 score, pvalue, _ = coint(df_pair['GS'], df_pair['MS'])
26
27 print(f"Testing Pair: {pair_tickers}")
28 print(f"t-statistic: {score:.4f}")
29 print(f"p-value: {pvalue:.4f}")
30 print(f"critical values:")
31 print(f"  1%: {crit_value[0]:.4f}")
32 print(f"  5%: {crit_value[1]:.4f}")
33 print(f" 10%: {crit_value[2]:.4f}")
34
35 if pvalue < 0.05:
36     print(">> RESULT: Cointegration Detected! (The Leash is holding)")
37 else:
38     print(">> RESULT: No Cointegration. (Relationship broken/spurious)")

```

Listing 1: Real-World Cointegration Verification

Results:

The empirical results indicate statistically significant cointegration between Goldman Sachs (GS) and Morgan Stanley (MS). The Augmented Dickey–Fuller test applied to the estimated spread yields a test statistic of -3.5678 and a p-value of 0.0268 , allowing the null hypothesis of a unit root to be rejected at the 5% significance level. This suggests that the spread between the two assets is stationary and exhibits mean-reverting behaviour. From a trading perspective, this implies the existence of a stable long-run equilibrium linking the prices of GS and MS, consistent with their shared exposure to the investment banking sector. Deviations from this equilibrium are therefore expected to be temporary rather than persistent, providing a potential basis for relative-value

```

Testing Pair: ['GS', 'MS']
t-statistic: -3.5678
p-value:      0.0268
critical values:
  1%: -3.9054
  5%: -3.3411
 10%: -3.0479

```

Figure 1: Values Found For GS and MS

strategies. However, while the result supports the presence of a tradable relationship over the sample period, it also highlights the inherent concentration risk of pair-based approaches, reinforcing the motivation for extending the analysis to multi-asset baskets using more general cointegration frameworks.

3 Johansen Test for Basket Trading

While the Engle-Granger method is an excellent pedagogical tool for understanding the mechanics of mean reversion, it suffers from a significant limitation in practice: it is strictly bivariate. It can only assess the relationship between two variables at a time.

In modern high-frequency markets, betting on a single pair exposes the trader to substantial idiosyncratic risk. If "Asset A" suffers a catastrophic, company-specific event (e.g., an accounting scandal), the cointegration breaks, and the spread widens indefinitely. To mitigate this, institutional quants scale up to "Basket Trading," creating portfolios in which one asset is hedged against a weighted basket of peers.

To handle this complexity, we must graduate from simple regression to the Johansen Test.

3.1 From Regression to VECM

When dealing with a portfolio of three or more assets (e.g., trading Exxon Mobil against a basket of Chevron, BP, and Shell), we cannot simply regress one against the others without losing information about the dynamic interactions between them. The Johansen Test solves this by moving beyond simple regression into a Vector Error Correction Model (VECM) [[2]]. Instead of a single equation, we model the changes in the entire vector of prices Y_t simultaneously:

$$\Delta Y_t = \Pi Y_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta Y_{t-i} + \epsilon_t$$

where:

- ΔY_t = change in the vector of prices Y_t
- Π = coefficient matrix of cointegrating relationships
- Γ = coefficient matrix of the lags of differenced variables of Y

- ϵ_t = error term

The critical component here is the matrix Π (Pi). In our previous "Drunk and the Dog" analogy, we only had one relationship to track. In a basket of N stocks, there might be multiple independent "leashes" binding them together. The matrix Π contains the information about these long-run relationships. The matrices Γ_i model the transient dynamics.

3.2 Eigenvalues and The Trace Statistic

The goal of the Johansen Test is to determine the Rank (r) of the matrix Π . The rank tells us exactly how many independent cointegrating relationships there are in our basket of stocks. If Rank (r) = 0: No cointegration exists. The assets are wandering randomly with no tether. If Rank (r) > 0: There are r stationary linear combinations.

To calculate this rank, we utilise Eigenvalues (λ). We treat the determination of the number of cointegrating vectors as a hypothesis testing problem using the Trace Statistic:

$$\lambda_{trace}(r) = -T \sum_{i=r+1}^n \ln(1 - \hat{\lambda}_i)$$

Intuitively, if the estimated eigenvalues ($\hat{\lambda}_i$) are significantly distinct from zero, it implies a strong cointegrating relationship. The Trace Statistic aggregates these values to test the null hypothesis that there are at most r cointegrating vectors.

3.3 Why This Matters: Robust Hedging

Understanding the rank allows us to construct a more resilient hedge. If we identify a basket of energy stocks with a rank of 1, we can find a single linear combination (a portfolio) that is stationary.

This allows us to trade the idiosyncratic noise of a single stock against the sector's systematic trend. Rather than betting that "Coke will outperform Pepsi" (a pair), we can bet that "Coke will revert to the mean of the entire Beverage Sector" (a basket). This diversification significantly reduces the risk of a single competitor's movement blowing up the trade.

3.4 Johansen Test Using Python:

While the Engle-Granger method suffices for pairs, analysing a portfolio requires a multivariate approach. We now implement the Johansen Test using the `coint_johansen` function from `statsmodels` to determine the rank of

```

1 # Necessary libraries
2 import yfinance as yf
3 import pandas as pd
4 import numpy as np
5 import statsmodels.api as sm
6 from statsmodels.tsa.stattools import coint_johansen
7
8
9 # Helper function to download and clean data
10 def get_prices(tickers, start_date='2021-01-01', end_date='2025-12-01'):
11     data = yf.download(tickers, start=start_date, end=end_date)['Close']
12     data = data.dropna() # Remove missing values
13     return data

```

```

14
15 # =====
16 # Johansen Test (6 Stocks) (Could be as many as wanted)
17 # =====
18
19 # Let's test the six Banks that have the most impact in the USA:
20 basket_tickers = ['JPM', 'MS', 'GS', 'BAC', 'WFC', 'C']
21 df_basket = get_prices(basket_tickers)
22
23 # Run Johansen Test
24 # det_order=0 implies we assume there is a constant term in the model
25 johansen_res = coint_johansen(df_basket, det_order=0, k_ar_diff=1)
26
27 # There are N (number of stocks) possible relationships (r=0, r=1, r=2...)
28 for r in range(len(basket_tickers)):
29     trace_stat = johansen_res.lr1[r]          # The test statistic
30     crit_val = johansen_res.cvt[r, 1]         # The 5% critical value (index 1)
31
32     # Logic: If Trace Stat > Critical Value, we REJECT the null hypothesis
33     if trace_stat > crit_val:
34         res = f"Reject H0 (At least {r+1} relationships exist)"
35     else:
36         res = f"Cannot Reject H0 (Likely only {r} relationships)"
37
38     print(f"r <= {r}      | {trace_stat:.4f}      | {crit_val:.4f}      | {res}")
39
40 # Eigenvalues help determine the strength of the relationship
41 print("\nEigenvalues of the matrix (Strength of Cointegration):")
42 print(johansen_res.eig)
43
44 print("\nINTERPRETATION:")
45 print("If we reject r<=0, it means there is at least 1 cointegrating relationship.")
46 print("If we reject r<=1, it means there are at least 2, and so on.")

```

Listing 2: Real-World Cointegration Verification Using Johansen Test

```

r <= 0 | 89.1784 | 95.7542 | Cannot Reject H0 (Likely only 0 relationships)
r <= 1 | 53.3018 | 69.8189 | Cannot Reject H0 (Likely only 1 relationships)
r <= 2 | 28.0519 | 47.8545 | Cannot Reject H0 (Likely only 2 relationships)
r <= 3 | 15.5642 | 29.7961 | Cannot Reject H0 (Likely only 3 relationships)
r <= 4 | 7.3227 | 15.4943 | Cannot Reject H0 (Likely only 4 relationships)
r <= 5 | 0.0275 | 3.8415 | Cannot Reject H0 (Likely only 5 relationships)

Eigenvalues of the matrix (Strength of Cointegration):
[2.38856961e-02 1.68708286e-02 8.37958620e-03 5.53817105e-03
 4.90384199e-03 1.85277308e-05]

INTERPRETATION:
If we reject r<=0, it means there is at least 1 cointegrating relationship.
If we reject r<=1, it means there are at least 2, and so on.

```

Figure 2: Results for Johansen Test

The Johansen Test applied to a basket of six major US banks (JPM, MS, GS, BAC, WFC, C) failed to identify a statistically significant cointegrating relationship. Specifically, in the test for rank $r \leq 0$ (the null hypothesis that zero cointegrating vectors exist), the calculated Trace Statistic of 89.18 fell below the 95% Critical Value of 95.75. Consequently, we are unable to reject the null

hypothesis. This statistical result is further corroborated by the near-zero eigenvalues, indicating that any potential mean-reverting signal is negligible. Economically, this implies that while these firms share the same sector, their individual price paths during this specific timeframe were driven closer by idiosyncratic risks rather than a shared fundamental tether. Therefore, a stationarity-based arbitrage strategy cannot be reliably constructed for this specific six-stock basket without modification.

4 From Math to Markets

Theory is useless without execution. Having mathematically confirmed a cointegrating relationship using the ADF or Johansen tests, we must now transform these statistical outputs into actionable trading signals.

4.1 Signal Generation: The Z-Score

The raw value of the spread (ϵ_t) is difficult to trade directly because its magnitude varies based on the price of the underlying assets. A spread of \$0.50 might be huge for a penny stock pair but negligible for a pair of tech giants. To standardise our signals, we normalise the spread into a Z-score. This tells us how many standard deviations the current spread is from its historical mean. [[5]]

$$Z_t = \frac{\epsilon_t - \mu_\epsilon}{\sigma_\epsilon}$$

Where:

- ϵ_t is the current spread value.
- μ_ϵ is the rolling mean of the spread.
- σ_ϵ is the rolling standard deviation.

This normalisation allows us to define systematic entry and exit rules:

- Entry: When $|Z_t| > 2$ (or another threshold like 1.5 or 3), the spread has stretched significantly. If $Z_t > 2$, the spread is "too high" (Asset A is overvalued relative to B), so we Short A / Long B. If $Z_t < -2$, we do the reverse.
- Exit: When Z_t returns to 0, the relationship has reverted to the mean, and we close the position to capture the profit.

4.2 Timing the Trade: The Ornstein-Uhlenbeck Process

A critical question in statistical arbitrage is, "How long will I be stuck in this trade?" If a mean reversion takes three years, the capital lockup makes the trade unviable. To estimate the expected holding period, we model the spread as an Ornstein-Uhlenbeck (OU) process, a stochastic process that describes mean-reverting behaviour:

$$d\epsilon_t = \theta(\mu - \epsilon_t)dt + \sigma dW_t$$

Where:

- θ (theta) represents the speed of mean reversion.
- μ represents the long-run mean of the error term
- σ represents the long-run volatility of the error term
- dW_t represents a change in a Wiener process (Brownian motion)

By fitting this differential equation to our residual series, we can calculate the Half-Life of the trade—the time expected for the spread to revert halfway back to its mean.

$$\text{Half Life} = \frac{\ln(2)}{\theta}$$

If the calculated Half-Life is extremely long (e.g., months), the "leash" is too loose, and the pair should likely be discarded in favour of faster-moving opportunities.

4.3 Risk Management: Structural Breaks

Finally, we must address the primary killer of stat-arb strategies: Structural Breaks. Cointegration is not a permanent law of physics; it is an economic observation that can change. A merger, a regulatory change, or a shift in a company's business model can permanently snap the "leash". If the spread widens beyond, say, 4 or 5 standard deviations ($Z > 5$), it is often statistically more likely that the relationship has broken rather than extended. In these cases, a hard stop-loss is essential to prevent betting on a mean reversion that will never come.

5 Application Using WorldQuant's Brain platform

Finally, we bridge the gap between academic theory and industry practice. While Python is an excellent environment for researching specific pairs, it is computationally heavy to scale across thousands of assets. Institutional platforms like WorldQuant Brain are designed for high-throughput testing, allowing us to generalise our cointegration logic across the entire market universe.

In this section, we will translate the mathematical formulas from Sections 1 and 2 into WorldQuant's expression-based syntax to build a mean-reversion "Alpha".

5.1 Translating the Math: Constructing the Signal

The trading strategy is evaluated using WorldQuant's BRAIN platform, which backtests alphas by mapping them into a systematic long-short portfolio over a predefined universe. In this framework, the strategy's signal determines cross-sectional position weights, with assets exhibiting stronger signals placed in the long book and weaker signals in the short book. The resulting portfolio is approximately dollar-neutral, so gains and losses reflect relative-value performance rather than overall market movements. This backtesting setup is well aligned with cointegration-based strategies, as it isolates the mean-reversion component of the spread while minimising exposure to broad market risk.

The alpha for this strategy is

$$- \text{ts_zscore}(\text{ts_sum}(\text{returns} - \text{group_mean}(\text{returns}, 1, \text{sector}), 60), 250).$$

The strategy first removes sector-wide effects by subtracting the sector mean return from each asset’s return, thereby isolating relative price deviations within economically related groups. These sector-adjusted returns are then aggregated over a 60-day window to capture persistent departures from a local equilibrium. The resulting series is normalised using a 250-day time-series z-score, expressing deviations relative to their historical variability. The negative sign induces a contrarian positioning, allocating long positions to assets that have underperformed their sector and short positions to those that have outperformed. Interpreted through the lens of statistical arbitrage, the signal exploits mean reversion in relative prices while remaining largely insulated from broader market and sector trends.

5.2 Results

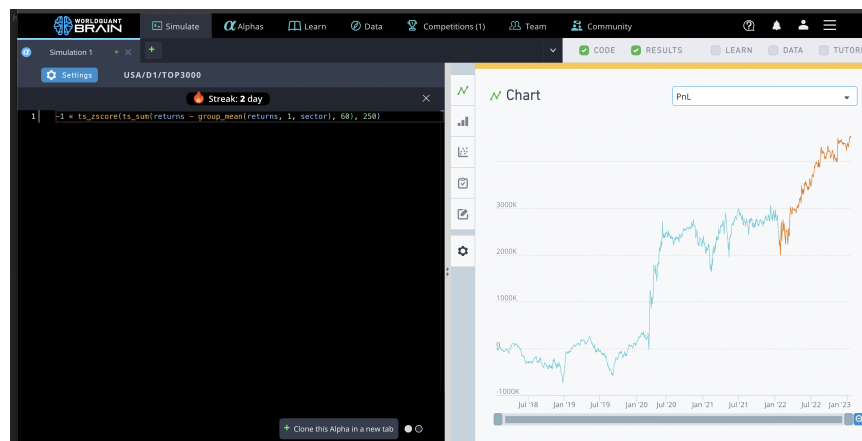


Figure 3: Brain platform results

The backtest results indicate that the strategy captures a persistent mean-reversion effect when implemented within a systematic long–short portfolio. The cumulative PnL exhibits a stable upward trend, consistent with repeatedly trading relative price deviations rather than directional market movements. Mathematically, this behaviour aligns with the interpretation of the signal as exploiting approximately stationary, sector-adjusted spreads.

The observed turnover of roughly 12% reflects the medium-horizon structure imposed by the 60-day aggregation and 250-day normalisation windows, indicating that the strategy responds to sustained mispricings rather than short-term noise. The moderate Sharpe ratio should be interpreted in the context of the dollar-neutral long–short framework, where returns are generated solely from cross-sectional dispersion. By construction, this limits absolute risk-adjusted performance while enhancing robustness across market regimes. Overall, the results are consistent with a risk-controlled statistical arbitrage strategy designed to extract mean reversion in relative price relationships.

It is important to note that due to computational constraints on high-frequency platforms, this Alpha implementation relies on a simplified sector-neutral mean reversion approach rather than a full VECM estimation. While strictly distinct from cointegration, this method acts as a

computationally efficient proxy by assuming the sector index acts as the 'integrating' tether for the individual stock.

6 Conclusion

This exploration has traversed the critical distinction between correlation and cointegration, moving from the intuitive "Drunk and the Dog" analogy to the rigorous mechanics of the Engle-Granger and Johansen tests. We established that while bivariate pairs trading offers a pedagogical foundation, modern statistical arbitrage requires the robustness of multi-asset baskets to mitigate idiosyncratic risk. By translating abstract econometric concepts such as stationarity and the Ornstein-Uhlenbeck process into actionable Z-score signals on platforms like WorldQuant Brain, we bridged the gap between theoretical math and practical execution. Ultimately, the durability of a quantitative strategy lies not in predicting random walks but in identifying and capitalising on the fundamental economic tethers that ensure mean reversion in a chaotic market.

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